

GENTS: Unified Generative Modeling for Multimodal Time Series Analysis

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Abstract

Modeling multimodal time series has emerged as an important research direction, aiming to incorporate auxiliary modalities such as textual descriptions to enhance temporal understanding. However, existing methods often treat external modalities as supplementary features, limiting their applicability to specific tasks and resulting in high costs in terms of design, training, and maintenance. In this paper, we propose GENTS, a unified generative framework for multimodal time series analysis that integrates multiple downstream tasks within a single model. GENTS is jointly trained for text-conditioned time series generation and multimodal forecasting, enabling effective modeling of the joint distribution between time series and external modalities. This unified formulation supports diverse applications, including conditional generation, forecasting, and time series editing. Moreover, by incorporating efficient time series captioning, GENTS alleviates the challenge of multimodal data scarcity. Extensive experiments on diverse real-world datasets demonstrate the effectiveness and generality of our approach across multiple tasks.

CCS Concepts

• Mathematics of computing → Time series analysis.

Keywords

multimodal time series, time series forecasting, conditional generation, time series editing, diffusion models

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1 Introduction

Time series are prevalent across real-world applications, including energy [23], climate [8, 20, 21], healthcare [3, 11, 15], and finance [7]. However, they rarely appear in isolation and are often accompanied by auxiliary modalities. For instance, a patient's physiological signals may be paired with diagnostic reports [42], or climate indicators may be supplemented by expert summaries [47]. These multimodal signals not only provide complementary context for the interpretation of time series, but also help uncover the underlying factors driving time series generation, enabling more effective analysis. As a result, more research focuses on integrating auxiliary modalities to enhance time series analysis, and recent surveys further position multimodal time series analysis as an emerging direction shaped by cross-modal fusion and alignment [17].

Multimodal time series analysis explores leveraging external modality in various downstream tasks. Forecasting is most widely studied [6, 9, 30, 31], where textual descriptions are commonly used to improve prediction accuracy. Another direction addresses time series generation [24, 33], conditioning the generation process on meta-information such as categories or attributes. Beyond generation, time series editing [19] seeks to modify existing time series in response to newly provided information.

While prior studies have demonstrated the effectiveness of incorporating multimodal data for time series analysis, the way to leverage the external modality information with consideration of time series remains an open problem. Existing works [29–31] often treat external modalities merely as supplementary features, such as domain-specific metadata, without modeling the joint distribution between time series and auxiliary modalities. Moreover, these approaches typically rely on task-specific architectures due to differences in modeling design and data formats, limiting their generalizability and the full potential of multimodal data.

To address the above limitations, we propose GENTS, a unified generative framework that captures the joint distribution of time series and auxiliary modalities, specifically textual descriptions¹. The term *unified* is used in a precise context: it refers to a single backbone and training pipeline that supports a variety of time series tasks, as opposed to relying on separate task-specific models or repeated retraining. We introduce a unified generative paradigm

¹Without loss of generality, we focus on textual information as the external modality for time series analysis.

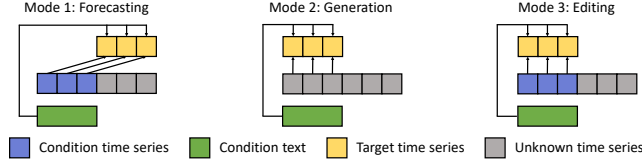


Figure 1: The unified generative paradigm of GENTS under different tasks. Forecasting is to generate future time series conditioned on historical ones and textual description, while generation (editing) aims at producing a time series based on the given textual information (and an original time series for modification).

that is applicable across diverse time series tasks. As shown in Fig. 1, each task in this framework involves modeling the target time series distribution conditioned on multimodal inputs, which may include time series segments, textual descriptions, or both. GENTS facilitates seamless adaptation to three downstream tasks within a unified generative framework, thereby reducing the costs of model design, training, and maintenance compared to task-specific approaches. Furthermore, it boosts overall performance through joint training.

In addition, multimodal modeling requires large amounts of paired multimodal data, yet high-quality pairs like time series with textual descriptions are scarce. Early attempts used coarse domain knowledge [18, 35], while others relied on online reports [31] or expert annotations [10], which are either weakly correlated or costly to scale. To address this, we propose a domain-agnostic captioning approach that integrates knowledge from both historical time series and external tools. This approach efficiently provides paired multimodal information, even in the absence of expert knowledge.

Our contributions are concluded as follows:

- We introduce GENTS, a novel generative modeling method that models joint distribution of multimodal time series data, and unifies multiple tasks of time series analysis into a single framework, which is effective and adaptable in various downstream tasks.
- Our method incorporates an automated time series captioning method, which significantly alleviate the scarcity issue of multimodal time series and eventually improves modeling.
- We constructed a comprehensive benchmark for generative modeling on various time series tasks, and proved the superiority of the unified framework compared with the task-specific methods.

2 Related Work

2.1 Multimodal Time Series Analysis

Many recent studies and surveys [17, 31, 51] have shown that numerical values in time series data alone provide limited information, while incorporating additional multimodal data offers valuable complementary insights. Various modalities have been explored to enhance time series analysis. Time-VLM [52] transforms time series into images and leverages the pretrained vision-language models (VLMs). Frequency has also been considered as an alternative perspective for interpreting time series, as ContextTST [13] models the frequency to enhance model generalization. Among all modalities, text stands out as the most intuitive for humans to

understand time series and is widely adopted in multimodal time series modeling. In our work, we take text modality as external information and explore multimodal time series analysis.

Previous literature has studied incorporating texts into time series modeling. In forecasting, several works introduce domain-level descriptions [32, 35] or sample-specific reports from the Internet [31, 48] as supplementary inputs to improve forecasting. However, such text often lacks both high relevance and semantic alignment with the corresponding time series. To address this limitation, some studies [25, 28, 41] attempt to interpret time series using human language, relying on high-quality, paired datasets of text and time series to train captioning models. Nevertheless, these methods tend to be dataset-specific and lack generalizability across domains. In time series generation, several works utilize metadata to guide the generation of time series [27, 33] or to modify existing ones [19]. Although metadata—often a simplified form of text—can be useful, it typically has a rigid structure and is insufficient for fine-grained control. Recent efforts [16, 18] aim to harness the capabilities of large language models (LLMs) by learning unified representations of both time series and text. These advancements motivate us to further explore the joint modeling of these two modalities.

Beyond time-series-specific works, our motivation is also connected to the broader multimedia literature on captioning, where language is treated as a semantic bridge rather than a literal restatement of low-level signals. For example, M&M TGM [2] uses multimodal latent topics to guide video captioning, and MA-LSTM [46] emphasizes that different modalities contribute differently during sentence generation. These works support that aligned language can serve as a bridge for cross-modal understanding and generation.

2.2 Time Series Modeling

Time series modeling can be generally categorized into discriminative and generative modeling, each serving different purposes depending on the task requirement. In forecasting, discriminative modeling has been the dominant approach. Early deep learning methods [34, 50] and foundation models [1] treat forecasting as a supervised regression problem, directly mapping historical sequences to future numerical values. Recent advancements have introduced generative modeling as an alternative framework for forecasting [39, 49], where models learn the probability distribution of future sequences conditioned on history, enabling uncertainty estimation and diverse forecasts. In parallel, diffusion has also been explored beyond direct generation. TSDE [37] studies diffusion-based self-supervised time series representation learning with an imputation-interpolation-forecasting mask, showing that diffusion can be used to learn transferable temporal representations rather than only task-specific predictors.

In generation and editing, generative modeling has become the standard paradigm. These tasks typically emphasize diversity, flexibility, and control over the generated outputs. Representative works include generating time series under specific constraints [5], discrete metadata [24, 27, 33], and modifying existing sequences based on attributes [19]. Addressing diverse time series tasks often requires task-specific architectures and objectives. Recent work [22] explores unified frameworks via unimodal unconditional generative pre-training, while TSDE [37] focuses on diffusion-based

representation learning. In contrast, our goal is a unified multimodal conditional generative framework that directly models the joint distribution of time series and text, and supports forecasting, generation, and editing within one backbone.

3 GENTS: Unified Generative Modeling for Time Series

In this work, we propose GENTS, a unified generative modeling framework designed to learn the joint distribution of time series and textual modality. We begin by formulating the multimodal generative modeling in Sec. 3.1 and introducing the backbone of our approach. Sec. 3.2 presents our unified modeling paradigm, followed by Sec. 3.3, where we describe how GENTS is adapted to a range of multimodal time series tasks. To address the challenge of limited paired data, we further propose a domain-agnostic time series captioning method in Sec. 3.4.

3.1 Time Series Generative Modeling

Given pairs of time series $\mathbf{x}$ and multimodal conditions $\mathbf{c}$ drawn from the joint distribution $q(\mathbf{x}, \mathbf{c})$, our objective is to learn a generative model $p_\theta(\mathbf{x}, \mathbf{c})$ that closely approximates the true joint distribution $q(\mathbf{x}, \mathbf{c})$. With condition $\mathbf{c}$ sampled from the real distribution $q(\mathbf{c})$, we have $p_\theta(\mathbf{x}, \mathbf{c}) = p_\theta(\mathbf{x}|\mathbf{c})q(\mathbf{c})$. To this end, we adopt the diffusion model [12] as the backbone of our generative framework, owing to its stability and effectiveness in capturing complex distributions, along with acceptable efficiency, which together make it well-suited for unifying multiple time series tasks with multimodal data.

During the training phase of the diffusion model, the forward process gradually adds noise to the original data sample $\mathbf{x}_0 = \mathbf{x}$ through a Gaussian Markov transition²:

$$\begin{aligned} q(\mathbf{x}_{1:T}|\mathbf{x}_0) &= \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}), \\ q(\mathbf{x}_t|\mathbf{x}_{t-1}) &= \mathcal{N}(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}), \end{aligned} \quad (1)$$

where $\{\beta_t\}_{t=1}^T$ are the predetermined variance schedule. Sampling of $\mathbf{x}_t$ has the closed-form written as $\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_0 + \sqrt{1-\alpha_t}\epsilon$, where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ and $\alpha_t := \prod_{s=1}^t (1 - \beta_s)$. To recover the original data $\mathbf{x}_0$ from the noisy version $\mathbf{x}_t$, a conditional noise estimator $\epsilon_\theta(\mathbf{x}_t, t, \mathbf{c})$ is trained to estimate the noise added to $\mathbf{x}_0$. The objective function can be written as:

$$\min_{\theta} \mathcal{L}(\mathbf{x}_0) = \min_{\theta} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}), t \sim \mathcal{U}(1, T)} \|\epsilon - \epsilon_\theta(\mathbf{x}_t, t, \mathbf{c})\|_2^2. \quad (2)$$

During the inference phase, given a condition $\mathbf{c}$, the sample $\hat{\mathbf{x}}_0$ can be generated from a gaussian noise $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$ through multiple denoising steps, which utilize the noise estimator $\epsilon_\theta(\mathbf{x}_t, t, \mathbf{c})$ with the deterministic Denoising Diffusion Implicit Model (DDIM) [38] sampler:

$$\begin{aligned} \hat{\mathbf{x}}_{t-1} &= \frac{\sqrt{\alpha_{t-1}}}{\sqrt{\alpha_t}} (\hat{\mathbf{x}}_t - \sqrt{1-\alpha_t}\epsilon_\theta(\hat{\mathbf{x}}_t, t, \mathbf{c})) \\ &\quad + \sqrt{1-\alpha_{t-1}}\epsilon_\theta(\hat{\mathbf{x}}_t, t, \mathbf{c}). \end{aligned} \quad (3)$$

The diffusion model has been widely adopted in time series modeling due to its stable and powerful distribution modeling capabilities [12]. However, its application has largely been limited to

² $\mathbf{x}$ and $\mathbf{x}_0$ are interchangeable in this paper.

single-task scenarios, such as imputation [40], forecasting [49], generation [33], and editing [19]. We argue that the full potential of diffusion models in capturing complex data distributions remains underexplored. Therefore, in the following sections, we investigate how diffusion models can be leveraged for universal generative modeling, aiming to construct a unified generative framework supporting multiple time series tasks.

3.2 Unified Generative Modeling

We formulate various time series tasks as *masked-conditional generation* problems and develop a unified generative framework based on diffusion models [12]. The central challenge in building such a unified model lies in standardizing the input-output interface across tasks, while allowing the model to distinguish among task-specific settings.

Masked-Conditional Generation. We introduce some notations below to support our formulation:

- $\mathbf{x} \in \mathbb{R}^L$: the original time series sequence of length L .
- $\mathbf{c}_{\text{text}} \in \mathbb{N}^W$: the textual description of the time series, where W is the number of tokens.
- $\mathbf{m}_{\text{cond}} \in \{0, 1\}^L$: a binary mask indicating conditional elements of $\mathbf{x}$. A value of one marks elements used as conditions while others keep zero.
- $\mathbf{m}_{\text{tgt}} \in \{0, 1\}^L$: a binary mask indicating target elements to be generated.

As illustrated in Fig. 1, the data formats vary across different tasks. Specifically, forecasting generates future time series conditioned on historical observations and textual descriptions, while generation and editing aim to produce or modify a time series based on textual information (and an original series for editing). Despite their differences, these tasks share a common structure: chronologically aligned condition and target segments of the time series, accompanied by corresponding textual descriptions. Thus, as shown in Fig. 2, we separately regard the time series $\mathbf{x}$ as $\mathbf{x}_{\text{cond}}$ and $\mathbf{x}_{\text{tgt}}$ using the condition mask $\mathbf{m}_{\text{cond}}$ and target mask $\mathbf{m}_{\text{tgt}}$, respectively.

Under our unified generative paradigm, we instantiate the time series $\mathbf{x}$ and multimodal condition $\mathbf{c}$ from the general diffusion formulation (Sec. 3.1) as follows: The target $\mathbf{x}_{\text{tgt}}$ represents the portion to be generated; and the multimodal condition is defined as $\{\mathbf{x}_{\text{cond}}, \mathbf{c}_{\text{text}}\}$. Accordingly, the generation task is formulated as learning the conditional distribution $p_\theta(\mathbf{x}_{\text{tgt}} | \mathbf{x}_{\text{cond}}, \mathbf{c}_{\text{text}})$. By explicitly utilizing condition and target masks, our proposed framework flexibly distinguishes between conditioning inputs and generation targets across a wide range of tasks, enabling a unified approach to modeling the joint distribution of multimodal time series data.

Unified I/O. The input $\mathbf{x}_t$ of the unified diffusion model for training is constructed as follows:

$$\begin{aligned} \mathbf{x}_t &= \mathbf{x}_{\text{cond}} + \mathbf{x}_{\text{tgt}} \\ &= \mathbf{m}_{\text{cond}} \circ \mathbf{x}_0 + \mathbf{m}_{\text{tgt}} \circ [\sqrt{\alpha_t}\mathbf{x}_0 + \sqrt{1-\alpha_t}\epsilon], \end{aligned} \quad (4)$$

where α_t and ϵ are defined in Sec. 3.1. Specifically, $\mathbf{x}_t$ consists of three parts: (i) the condition segment corresponding to the index set: $\{i | \mathbf{m}_{\text{cond}}^i = 1\}$; (ii) the target segment for generation corresponding to the index set: $\{i | \mathbf{m}_{\text{tgt}}^i = 1\}$; (iii) the unknown segment corresponding to the index set: $\{i | \mathbf{m}_{\text{cond}}^i = 0, \mathbf{m}_{\text{tgt}}^i = 0\}$. In general, the input to the unified generative model includes the task type

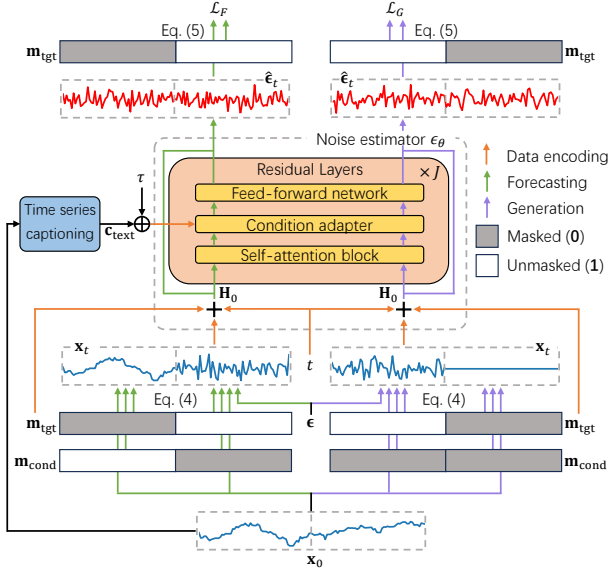


Figure 2: The overall pipeline of GENTS, including the unified generative modeling framework and domain-agnostic captioning.

τ , the time series sequence $\mathbf{x}_t$, the textual description $\mathbf{c}$, and the diffusion step t . The output is the estimated noise $\hat{\epsilon}_t \in \mathbb{R}^L$, where $\hat{\epsilon}_t = \epsilon_\theta(\mathbf{x}_t, \mathbf{c}_{\text{text}}, \tau, t)$.

Unified Optimization Objective. Our optimization objectives are primarily divided into two categories: forecasting the future and generating the history. Within the diffusion model framework, both tasks are uniformly formulated as the estimation of noise in the target generation segment, specified by the mask $\mathbf{m}_{\text{tgt}}$. Different masks are employed to distinguish between the task types:

$$\mathcal{L} = \sum_{\tau \in \{F, G\}} \lambda_\tau \times \mathcal{L}_\tau, \quad (5)$$

$$\mathcal{L}_\tau = \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}), t \sim \mathcal{U}(1, T)} \|\mathbf{m}_{\text{tgt}} \circ [\epsilon_\theta(\mathbf{x}_t, \mathbf{c}_{\text{text}}, \tau, t) - \epsilon]\|_2^2,$$

where $\circ$ is Hadamard product, F and G denote forecasting and generation, respectively, and λ_τ is a hyperparameter used to balance the loss contribution from each task. We will further discuss the variations in the condition mask $\mathbf{m}_{\text{cond}}$ and the target generation mask $\mathbf{m}_{\text{tgt}}$ in Sec. 3.3. The experimental results in Sec. 4.4 demonstrate that unified training with a balanced loss weight improves the performance of specific tasks.

Unified Model Architecture. We design our model architecture with reference to [36], which effectively models interactions between different modalities. The inputs are first transformed into embeddings through a series of encoders. As shown in Fig. 2, the noisy time series sequence $\mathbf{x}_t$ and the generation target mask $\mathbf{m}_{\text{tgt}}$ are encoded into the patch sequence $\mathbf{P} \in \mathbb{R}^{N \times D}$ and the patch mask embedding $\mathbf{m}_p \in \mathbb{R}^{N \times D}$, respectively, where $N = \lfloor \frac{L-P}{P} \rfloor$, P is the patch size. The diffusion step t is embedded into $\mathbf{e}_t \in \mathbb{R}^D$. To distinguish different tasks, we prepend the task type token τ to the textual condition $\mathbf{c}_{\text{text}}$ and use a text encoder to obtain the embedding $\mathbf{e}_c \in \mathbb{R}^D$. Unless otherwise specified, the text encoder is

trained from scratch together with the diffusion model, rather than initialized from a separately pretrained language model.

We model the influence of the time series condition and the textual condition through two mechanisms: self-attention and layer normalization adapter. The unified noise estimator consists of J layers, each containing an adapter, a self-attention block, and a feed-forward network. For the j -th layer, the computation is:

$$\begin{aligned} \mathbf{z}_1, \mathbf{z}_2 &= \text{MLP}(\mathbf{e}_c), \\ \tilde{\mathbf{H}}^j &= \mathbf{H}^j + \mathbf{z}_1, \\ \mathbf{H}^j &= \text{FFN}(\text{SA}(\tilde{\mathbf{H}}^{j-1}) + \mathbf{z}_2) + \mathbf{H}^{j-1}, \end{aligned} \quad (6)$$

where $\mathbf{H}^0 = \mathbf{P} + \mathbf{m}_p + \mathbf{e}_t$, MLP denotes the adapter module, SA is the self-attention block, and FFN is the feed-forward network. The interaction between the textual condition $\mathbf{c}_{\text{text}}$ and the noisy time series $\mathbf{x}_t$ is captured via the adapter MLP, while the interaction between the condition segment and target segment in $\mathbf{x}_t$ is modeled through the self-attention block SA . Finally, the output from the last layer $\mathbf{H}^J$ is passed through a patch decoder to produce the estimated noise $\hat{\epsilon}_t$. For more details of the unified noise estimator, please refer to Appendix ??.

Compared to the existing unimodal approach [22], which unifies different tasks through unconditional generation combined with self-guidance, we introduce a unified generative modeling framework grounded in the masked-conditional generation paradigm to better address the challenges inherent in multimodal time series tasks. By leveraging the condition adapter and masked self-attention mechanism, our method enables more flexible processing of multimodal information, thereby improving its applicability across a broader range of scenarios.

3.3 Tackling Multiple Tasks of Time Series Analysis

Following the unified generative modeling framework, GENTS addresses a variety of time series tasks under a common paradigm, to generate time series conditioned on multimodal inputs. As outlined in Sec. 3.2, different combinations of condition and target masks are employed to distinguish between tasks within this unified formulation. To support this, we partition each time series sequence of length L into two segments: a history segment of length L_h and a future segment of length L_f , with $L = L_h + L_f$. This segmentation provides a consistent structure across tasks, enabling flexible condition-target configurations based on the specific task requirements.

Forecasting. Multimodal time series forecasting aims to predict future values based on historical sequences and texts, while strictly avoiding the use of future information to prevent data leakage. We construct the condition mask as $\mathbf{m}_{\text{cond}} = [1^{L_h}, 0^{L_f}]$ and the target mask as $\mathbf{m}_{\text{tgt}} = [0^{L_h}, 1^{L_f}]$. The input sequence is defined as $\mathbf{x}_t = \mathbf{m}_{\text{cond}} \circ \mathbf{x}_0 + \mathbf{m}_{\text{tgt}} \circ \hat{\mathbf{x}}_{t+1}$, where $\hat{\mathbf{x}}_T \sim \mathcal{N}(0, \mathbf{I})$, T is the diffusion step number. A task-specific token “< forecasting >” is prepended to the texts $\mathbf{c}_{\text{text}}$ to indicate the forecasting task and provide contextual guidance.

Generation. Time series generation [33] focuses on synthesizing a sequence solely based on a text. The condition mask is set as $\mathbf{m}_{\text{cond}} = \mathbf{0}^L$ and the generation target mask as $\mathbf{m}_{\text{tgt}} = [1^{L_h}, 0^{L_f}]$. The input time series sequence is constructed as $\mathbf{x}_t = \mathbf{m}_{\text{cond}} \circ \mathbf{x}_0 + \mathbf{m}_{\text{tgt}} \circ \hat{\mathbf{x}}_{t+1}$, where $\hat{\mathbf{x}}_T \sim \mathcal{N}(0, \mathbf{I})$, T is the diffusion step.

Table 1: Summary of attributes extracted using tsfresh [4].

Attribute Type	Attribute Name	Description
Statistical	Mean/Std/Min/Max	The mean, standard deviation, minimum and maximum value of the time series.
	Skewness/Kurtosis	Measures the asymmetry and sharpness of the time series value distribution.
	Frequency	Identifies the dominant periodicity in the frequency domain.
Morphological	Overall/Local Trend	Captures the trend slope of the entire time series and each segment.
	Number of Peaks	Counts the number of local maxima within each segment.
	Complexity	Quantifies the roughness of each segment.

type token “<generation>” is prepended to the textual description $\mathbf{c}_{\text{text}}$, which provides information about the historical segment.

Editing. Time series editing [19] seeks to revise an existing sequence in accordance with new textual guidance. The editing capabilities of GENTS evolve naturally through joint training, without requiring additional training. It uses the same mask configuration and task token “< generation >” as in the generation task. However, during inference, rather than initializing from Gaussian noise like the generation, the input is constructed by perturbing the original sequence through the forward diffusion process, following [19]: $\mathbf{x}_t = \mathbf{m}_{\text{cond}} \circ \mathbf{x}_0 + \mathbf{m}_{\text{tgt}} \circ \hat{\mathbf{x}}_{t+1}$, where $\hat{\mathbf{x}}_T = \sqrt{\alpha_T} \mathbf{x}_0 + \sqrt{1 - \alpha_T} \epsilon$, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, T is the diffusion step number. The model then applies the learned noise estimator to denoise $\mathbf{x}_t$ conditioned on the updated textual input $\mathbf{c}_{\text{text}}$, yielding the edited output $\hat{\mathbf{x}}_0$.

3.4 Domain-Agnostic Time Series Captioning

To address the scarcity of paired time series and text, we propose a general framework for time series captioning inspired by how humans intuitively interpret unfamiliar time series: focusing on *statistical* and *morphological* attributes. Statistical attributes, such as mean, skewness, and kurtosis, summarize global distributional properties, while morphological attributes capture structural patterns like trends, peaks, and abrupt changes. Crucially, these attributes are domain-agnostic, offering a shared descriptive basis across diverse fields. For instance, both a temperature drop due to rainfall and a stock price decline from policy shifts can be described as a downward trend. This abstraction enables generalizable, informative captioning without relying on domain-specific knowledge.

In this section, we propose a general captioning method that extracts attributes from time series and converts them into texts using predefined templates. Our approach leverages tsfresh [4] as the attribute annotator, denoted by $\mathcal{T}$. We define a set of attributes to be extracted, with corresponding statistical and morphological parameters denoted by $\mathcal{P} = \{\mathcal{P}_{\text{sta}}, \mathcal{P}_{\text{mor}}\}$. The complete list of attributes is provided in Tab. 1. Given a time series $\mathbf{x}$, the attribute annotator $\mathcal{T}$ extracts the corresponding attribute set as follows:

$$\mathcal{A} = \mathcal{T}(\mathbf{x}, \{\mathcal{P}_{\text{sta}}, \mathcal{P}_{\text{mor}}\}), \quad (7)$$

where $\mathcal{A} = \{\mathcal{A}_i\}_{i=1}^M$ denotes the set of extracted attributes, and each $\mathcal{A}_i \in \mathbb{R}$ represents the value of the i -th attribute. These attributes are then discretized and mapped to natural language using predefined templates. To avoid overly rigid descriptions, we maintain multiple template variants for several attribute categories and randomly sample from them when constructing the final caption.

This captioning strategy offers two practical benefits. First, it alleviates the scarcity of paired multimodal data by providing aligned textual descriptions even for originally unimodal datasets. Second, it supplies fine-grained semantic conditions that are more expressive than coarse metadata, while remaining more stable and less hallucination-prone than free-form captions generated by large language models. Compared with existing time series captioning methods [25, 28, 41], which rely on training captioning models using high-quality text from specific domains, our approach enables cross-domain captioning without training. Furthermore, our approach leverages both the knowledge from historical sequences and external tools, offering a more informative and effective solution for aligned text generation across diverse domains. We provide more details about the text synthesis in Appendix ??.

4 Experiment

In this section, we present the experiment settings and the corresponding results, organized around the following research questions (RQs): **RQ1**: Is unified generative modeling effective for multiple downstream tasks? **RQ2**: Does the synthetic domain-agnostic text improve the performance of different tasks? **RQ3**: What’s the impact of the unified generative modeling in different tasks?

4.1 Experiment Setup

Datasets. We utilize two categories of datasets: (i) Multimodal datasets consisting of time series and manually collected textual descriptions. These include **TimeMMD-Env**, the largest subset of TimeMMD datasets [31], **Weather** dataset [47], and an original dataset **HongKong** [14] collected by ourselves. (ii) Unimodal datasets consisting of time series only, including **ETTm1** [53], **Exchange** [23], and **Traffic** [26]. More details regarding dataset statistics and the specific data pairing protocol for the editing task are provided in Appendix ??.

Evaluation Metrics. For the forecasting task, we report mean squared error (MSE) and mean absolute error (MAE). For the generation task, Joint Frechet Time Series Distance (**J-FTSD**) [33] is used to quantify the discrepancy between the distribution of generated and real data; Contrastive Time Series Text Pretraining (**CTTP**) score is incorporated to measure the similarity between the generated time series and the ground truth time series. For the editing task, we adopt the Log Ratio of Target-to-Source (**RaTS**) [19] and the **CTTP** score to evaluate the model’s ability to modify existing

Table 2: Averaged performance of generation, editing, and forecasting tasks on multimodal (left) and unimodal (right) datasets. The best results are shown in bold, and the second-best results are underlined. Arrows $\uparrow$ ($\downarrow$) indicate that higher (lower) values are better.

Task	Multimodal datasets						Unimodal datasets					
Generation	TimeMMD-Env		Weather		HongKong		ETTm1		Exchange		Traffic	
	$\downarrow$ JFTSD	$\uparrow$ CTTP	$\downarrow$ JFTSD	$\uparrow$ CTTP	$\downarrow$ JFTSD	$\uparrow$ CTTP	$\downarrow$ JFTSD	$\uparrow$ CTTP	$\downarrow$ JFTSD	$\uparrow$ CTTP	$\downarrow$ JFTSD	$\uparrow$ CTTP
TimeWeaver [33]	421.145	79.728	291.369	55.809	458.184	120.509	40.281	93.554	78.282	85.068	140.031	55.520
TEdit [19]	423.662	76.808	290.678	54.996	457.276	120.956	30.761	108.750	80.606	82.621	141.305	55.097
GENTS (ours)	166.379	83.687	1.320	115.251	23.516	140.052	27.227	127.575	45.801	180.377	178.937	44.942
Editing	TimeMMD-Env		Weather		HongKong		ETTm1		Exchange		Traffic	
	$\uparrow$ RaTS	$\uparrow$ CTTP	$\uparrow$ RaTS	$\uparrow$ CTTP	$\uparrow$ RaTS	$\uparrow$ CTTP	$\uparrow$ RaTS	$\uparrow$ CTTP	$\uparrow$ RaTS	$\uparrow$ CTTP	$\uparrow$ RaTS	$\uparrow$ CTTP
TimeWeaver [33]	0.584	79.238	0.364	56.035	0.088	117.900	0.282	93.854	-1.411	88.967	-0.622	52.182
TEdit [19]	0.556	76.011	0.370	55.962	0.078	117.916	0.551	107.843	-1.577	83.941	-0.498	56.721
GENTS (ours)	1.451	84.059	3.071	115.281	0.121	140.376	0.795	128.247	0.796	180.166	4.412	43.434
Forecasting	TimeMMD-Env		Weather		HongKong		ETTm1		Exchange		Traffic	
	$\downarrow$ MAE	$\downarrow$ MSE	$\downarrow$ MAE	$\downarrow$ MSE	$\downarrow$ MAE	$\downarrow$ MSE	$\downarrow$ MAE	$\downarrow$ MSE	$\downarrow$ MAE	$\downarrow$ MSE	$\downarrow$ MAE	$\downarrow$ MSE
DLinear [50]	0.627	0.595	0.304	0.205	0.922	1.433	0.722	1.175	0.253	0.129	0.575	0.511
Autoformer [45]	0.835	1.218	0.444	0.377	1.833	5.505	1.257	2.815	0.949	1.795	2.942	12.350
TimesNet [44]	0.538	0.556	0.288	0.207	0.786	1.108	0.790	1.380	0.183	<u>0.066</u>	0.486	0.381
Diffusion-TS [49]	0.532	0.587	0.421	0.406	0.820	1.233	0.746	1.107	0.323	0.198	0.575	0.610
Moirai [43]	0.736	1.629	0.349	0.280	0.888	1.458	1.422	1.751	0.180	0.069	1.157	2.109
Chronos [1]	0.634	0.790	0.292	0.217	0.798	1.192	0.771	1.277	<u>0.176</u>	0.065	0.656	0.728
TimeCMA [30]	0.534	0.538	0.361	0.270	0.780	1.072	0.791	1.339	0.185	0.065	0.926	1.072
TimeMMD [31]	0.506	0.468	0.326	0.203	0.754	1.017	<u>0.724</u>	1.235	0.174	<u>0.066</u>	<u>0.465</u>	0.423
IATSF [48]	0.510	0.514	<u>0.267</u>	<u>0.169</u>	0.735	0.974	0.767	1.245	0.166	<u>0.066</u>	0.916	1.036
GENTS (ours)	<u>0.509</u>	<u>0.508</u>	0.221	0.139	<u>0.743</u>	<u>1.003</u>	0.722	<u>1.168</u>	0.182	0.065	0.456	<u>0.420</u>

time series according to new textual descriptions. Further details on the evaluation metrics are provided in Appendix ??.

Baselines. We compared our method with different baselines for generation, editing, and forecasting tasks. For generation and editing, the metadata-based method **TimeWeaver** [33], **TEdit** [19] are considered. For forecasting, we consider both the unimodal and multimodal methods. The unimodal models include the MLP method **DLinear** [50], CNN method **TimesNet** [44], transformer method **Autoformer** [45], diffusion method **Diffusion-TS** [49], and foundation models **Moirai** [43] and **Chronos** [1]. The multimodal models include **TimeCMA** [30] and **TimeMMD** [31], **IATSF** [48].

Setting. GENTS is implemented as a unified multi-task framework trained from scratch on each individual dataset, rather than a pre-trained foundation model. Specifically, for the main results reported in Tab. 2, a single GENTS model is trained jointly on all relevant tasks using only the training split of the target dataset. It is then evaluated on the corresponding test split of the same dataset.

We adopt different configurations of history length, forecast horizon, and generation length for different datasets. For the

TimeMMD-Env, ETTm1, Exchange, and Traffic datasets, the setting is (48, {48, 96, 144}, 48). The configuration is (28, {28, 56, 84}, 28) for HongKong dataset and (36, {36, 72, 108}, 36) for Weather dataset. On unimodal datasets, GENTS utilizes only the synthetic domain-agnostic text. On multimodal datasets, GENTS additionally incorporates the real textual descriptions provided in the datasets. We set the loss weight ratio of $\lambda_F : \lambda_G$ to 2 : 1, the number of epochs to 700, the batch size to 1024, and the learning rate to 1×10^{-3} across all datasets. All experiments were conducted on a single NVIDIA-A40 GPU running three times with different random seeds.

4.2 Quantitative Results

In this section, we quantitatively evaluate the generation, editing, and forecasting performance of GENTS and baselines across all unimodal and multimodal datasets. The results are presented in Tab. 2. It's important to note that, in the absence of textual descriptions provided by the datasets, TimeMMD functions as a unimodal model, similar to PatchTST. For a fair comparison, we exclude TimeMMD

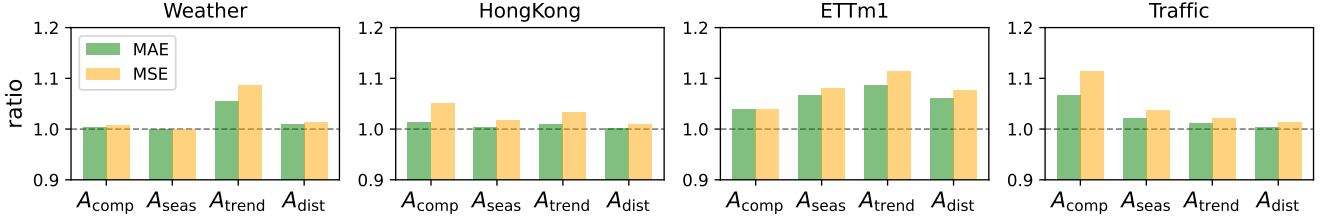


Figure 3: Error increase ratio of forecasting after *masking* the textual description of a certain class of attributes, where the higher ratio represents the larger impact on the forecasting performance. The masked attribute classes include $\mathcal{A}_{comp}$ (complexity-related), $\mathcal{A}_{seas}$ (season-related), $\mathcal{A}_{trend}$ (trend-related), and $\mathcal{A}_{dist}$ (distribution-related). The results are the average of different horizons on four datasets: Weather, HongKong, ETTm1, and Traffic.

from forecasting evaluations on unimodal datasets. Forecasting results are reported as averages across different horizons. Details for individual horizons forecasting, visual case studies demonstrating the editing and generation capabilities, and model efficiency are provided in Appendix ??.

Finding 1: *GENTS can effectively handle forecasting, generation, and editing within a single model, while achieving strong performance across all three tasks.* As shown in Tab. 2, GENTS performs competitively or superiorly across both multimodal and unimodal datasets, despite using one unified formulation rather than separate task-specific designs.

For generation and editing, GENTS consistently outperforms the baselines on most datasets. We attribute this advantage to its fine-grained textual conditioning and its explicit multimodal generative formulation, which allow the model to align generated time series more closely with semantic instructions than methods based on coarse metadata or attribute editing alone. The gains are particularly evident on multimodal datasets, where richer text and time-series alignment can be exploited directly. For forecasting, GENTS remains highly competitive with strong task-specific and foundation-model baselines. This suggests that the unified framework does not sacrifice forecasting quality, even though it is optimized jointly with other tasks. Instead, the shared generative formulation appears to preserve forecasting accuracy while broadening the range of supported applications.

Overall, these results answer **RQ1** that a unified multimodal generative model can support multiple downstream tasks effectively.

4.3 Ablation Study

In this section, we present an ablation study of GENTS to examine the impact of unified modeling and the use of synthetic domain-agnostic text. Without loss of generality, we select the HongKong and Traffic datasets, representing a multimodal and a unimodal dataset, respectively.

Finding 2: *Synthetic domain-agnostic text improves forecasting accuracy and provides fine-grained control for generation.* As shown in Tab. 3, removing the textual condition leads to a substantial degradation in both forecasting and generation performance on HongKong and Traffic. This result indicates that the synthesized text is not merely redundant side information; instead, it serves as a useful semantic summary of the underlying time series.

For forecasting, the text provides an additional perspective of the historical sequence by explicitly exposing global statistics and structural patterns that may be difficult to infer from raw values alone. For generation, the text acts as a controllable semantic condition that constrains the target sequence more precisely than unconditional generation. These observations answer **RQ2** and confirm that domain-agnostic text is beneficial not only for multimodal datasets with natural language annotations, but also for originally unimodal datasets.

Table 3: The ablation study on HongKong and Traffic datasets. $\mathcal{L}_F$ is the forecasting loss, $\mathcal{L}_G$ is the generation loss, and C_{text} is the text condition. Since evaluating forecasting and generation without the corresponding losses $\mathcal{L}_F$ and $\mathcal{L}_G$ is meaningless, we leave these results blank.

Method	Forecasting				Generation			
	HongKong		Traffic		HongKong		Traffic	
	↓MAE	↓MSE	↓MAE	↓MSE	↓JFTSD	↑CTTP	↓JFTSD	↑CTTP
GENTS	0.743	1.003	0.456	0.420	23.516	140.052	178.937	44.942
w/o $\mathcal{L}_F$	-	-	-	-	23.477	141.537	175.805	46.486
w/o $\mathcal{L}_G$	0.764	1.065	0.563	0.601	-	-	-	-
w/o C_{text}	0.909	1.533	0.570	0.605	36.876	112.978	211.512	21.248

Finding 3: *Unified training of forecasting and generation can enhance forecasting performance while preserving generation quality.* As shown in Tab. 3, jointly modeling forecasting and generation yields better forecasting performance than modeling forecasting alone, while maintaining comparable generation quality. This suggests that both tasks can be formulated as conditional probabilistic modeling problems, and that their joint optimization enables forecasting to benefit from shared representations. These findings address **RQ3**, demonstrating that unified modeling not only preserves individual task performance but can also enhance the performance of specific tasks. Moreover, we observe a trade-off between the two tasks, which we further analyze through a sensitivity study on the loss weight ratio $\lambda_F : \lambda_G$. Additional details are provided in Sec. 4.4.

4.4 Extended Analysis

Finding 4: *Statistical attributes are consistently beneficial for forecasting across all datasets, whereas morphological attributes contribute to varying degrees depending on the dataset.* To interpret

the impact of different attributes on forecasting, we mask the corresponding textual descriptions and compute the error-increase ratios: $\frac{MAE_{mask}}{MAE}$ and $\frac{MSE_{mask}}{MSE}$. We categorize all the attributes (detailed in Tab. 1) into 5 classes: value-statistical (Mean/Std/Min/Max), complexity-related (Complexity/Number of Peaks), season-related (Frequency), trend-related (Overall/Local Linear Trend), and distribution-related (Skewness/Kurtosis). The results show that value-statistical attributes have a substantial impact across all datasets, with masking leading to an error increase ranging from 28% to 98%. In contrast, the influence of the other attributes varies across datasets. As shown in Fig. 3, each attribute class significantly affects the forecasting performance in at least one dataset, demonstrating that all extracted attributes are useful for forecasting.

Table 4: The zero-shot forecasting performance of GENTS and GENTS without C_{text} . The model is trained on the dataset on the left of $\rightarrow$, and evaluated on the right of $\rightarrow$.

Method	Exchange $\rightarrow$ Traffic		ETTm1 $\rightarrow$ Traffic		Exchange $\rightarrow$ ETTm1	
	$\downarrow$ MAE	$\downarrow$ MSE	$\downarrow$ MAE	$\downarrow$ MSE	$\downarrow$ MAE	$\downarrow$ MSE
GENTS	1.206	2.230	1.034	1.515	0.985	2.112
GENTS w/o C_{text}	1.404	3.192	1.060	1.694	1.050	2.401

Finding 5: Synthetic domain-agnostic text can enhance zero-shot forecasting performance. Distinct from the supervised evaluation in Tab. 2, where models are trained and tested on the same dataset, we conduct a separate analysis to probe the cross-dataset generalization capability of our domain-agnostic text features. As shown in Tab. 4, models augmented with domain-agnostic text achieve better performance when performing zero-shot forecasting on unseen datasets. This improvement indicates that domain-agnostic text captures the statistical and morphological characteristics of time series, which often reflect common time series patterns shared across different datasets and thus improve cross-dataset transferability.

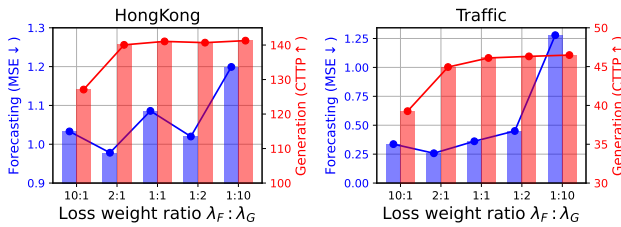


Figure 4: The sensitivity analysis of loss weight on HongKong (left) and Traffic (right) datasets. The loss weight ratio $\lambda_F : \lambda_G$ range in [10:1, 2:1, 1:1, 1:2, 1:10]. The further to the left, the higher the proportion of forecasting; otherwise, the higher the proportion of generation. MSE is reported for forecasting performance, CTTP is reported for generation performance.

Finding 6: Task performance is robust to the loss weight hyperparameter, and similar preferences generalize across datasets. We conduct a sensitivity analysis on the loss weight ratio $\lambda_F : \lambda_G$ to examine how the trade-off between forecasting and generation

affects the overall behavior of the unified model. As shown in Fig. 4, except under extreme ratios, both forecasting and generation performances remain relatively stable, demonstrating the robustness of our method to different optimization preferences. Furthermore, we observe that different datasets exhibit similar preferences for the loss weight ratio. The most balanced configuration for the two tasks is around $\lambda_F : \lambda_G = 2 : 1$, which we adopt consistently across all datasets.

Table 5: Human evaluation results on time series generation. Results are reported as average rank $\pm$ standard deviation, where lower is better.

Model	Fidelity Avg. Rank ($\downarrow$)	Instruction Alignment Avg. Rank ($\downarrow$)
TimeWeaver	2.033 $\pm$ 0.184	2.100 $\pm$ 0.082
TEdit	2.000 $\pm$ 0.308	2.267 $\pm$ 0.165
GENTS	1.967$\pm$0.409	1.633$\pm$0.155

Finding 7: GENTS achieves better human-perceived instruction alignment in time series generation. To complement the automatic metrics and further verify the effectiveness of GENTS, we conduct a human preference study focusing on fidelity and instruction alignment. We randomly sample 20 textual descriptions from the test sets and generate time series with TimeWeaver [33], TEdit [19], and GENTS. For each description, three evaluators are presented with the three anonymized outputs in randomized order and independently rank them from 1st to 3rd under the two criteria. This yields 60 rank scores per model and criterion, and we report the ranks in Tab. 5, where lower is better.

Although all methods obtain similar fidelity ranks with relatively large deviations, suggesting that visually judging the realism of generated time series is difficult for humans, instruction alignment is easier to distinguish. GENTS achieves the best alignment rank by a clear margin, showing that the unified generative framework better captures the semantic intent of textual descriptions.

5 Conclusion

The integration of multimodal information has become increasingly prominent in time series analysis. However, existing methods are often tailored to specific tasks, relying on specialized architectures and training strategies. Their scalability is further limited by the scarcity of large-scale multimodal datasets. In this work, we introduce GENTS, a novel unified multimodal generative framework based on diffusion models that supports multiple time series tasks, including forecasting, generation and editing, within a unified architecture, enabling flexible adaptation and reducing the development cost and complexity. To mitigate the shortage of paired multimodal data, we further propose a domain-agnostic captioning method that automatically generates textual descriptions from statistical and morphological attributes of time series. Experiments demonstrate that GENTS consistently outperforms baselines on different tasks.

Despite its strengths, GENTS still has limitations. Currently, it only generates time series outputs and does not yet support richer multimodal outputs or interactive reasoning. Nevertheless, we believe this work offers a promising step toward unified modeling of multimodal time series and can inspire future research on unified multimodal generative frameworks.

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